

SET perihelion derivation 14

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We derive the stationary spherical burden profile and its perihelion equation by requiring SET's physical-current law to agree with its source-surface relation across a continuous family of spherical sources. The radial conversion is left unknown until those requirements select it. The material action then gives the conserved energy, angular momentum and exact orbit equation, followed by the leading perihelion advance.

This reformulates the spherical construction of derivation 8c [1]. Its physical-gradient current and complete-screen conservation law are retained. Its separate carrier relation is recovered as a consequence. The selection also assumes a universal local response $H(A)$; it is a conditional result within that class, rather than a uniqueness theorem from the scalar invariant alone. The metric-completion action used in derivation 13 [2] is not an input to this route.

We consider an isolated, positive, nonrotating spherical source, a stationary vacuum exterior, and a test body whose backreaction is neglected. The derivation is restricted to the region where the held-clock capacity is positive. Source multipoles, other bodies and radiation reaction lie outside this calculation.

1 The invariant and the unresolved radial response

Start with the capacity invariant and its event-progression ratio [3]:

$$S^2 + V_{\text{rel}}^2 + c^2 \beta^2 = c^2, \beta = \frac{d\tau}{dT}. \quad (1)$$

Here T is the stationary reference time normalized to the unloaded clock at infinity, τ is the recorded time of the probe, and S is gravitational burden in speed units. Define the held-clock factor α and use A for its square:

$$A = \alpha^2 = 1 - \frac{S^2}{c^2}, 0 < A \leq 1. \quad (2)$$

Throughout this derivation A is dimensionless; a geometric screen area is written A_Σ . The coordinate r is areal radius, so $A_\Sigma(r) = 4\pi r^2$. In a static spherical description, write the physical travel norm as

$$V_{\text{rel}}^2 = \zeta^2 \left(\frac{dr}{dT} \right)^2 + r^2 \left(\frac{d\Omega}{dT} \right)^2, \zeta > 0. \quad (3)$$

The angular abbreviation is $d\Omega^2 = d\theta^2 + \sin^2\theta d\phi^2$. The factor ζ converts an areal-radius increment to radial physical distance. Leaving it unknown prevents the desired radial response from entering through a definition.

2 The response class and the physical source surface

Parameterize the radial conversion by one positive, continuous function of the local held-clock capacity:

$$\zeta^2 = \frac{H(A)}{A}, H(1) = 1. \quad (4)$$

The same function H applies to every member of the spherical source family. This class contains no independent dependence on source radius, density gradients, history or a further field. Spherical symmetry alone does not impose that restriction; it is part of the response class being tested. The unloaded normalization fixes the physical radial unit at infinity.

Multiplication of the invariant by dT^2 gives the general stationary interval in this class:

$$c^2 d\tau^2 = A c^2 dT^2 - \frac{H(A)}{A} dr^2 - r^2 d\Omega^2. \quad (5)$$

A held clock has $d\tau_{\text{hold}} = \alpha dT$, and a radial physical increment has $d\ell = \zeta dr$. Neither relation has selected $H = 1$. The same freedom is present if ζ is interpreted directly as the radial conversion in V_{rel} .

Now take a physical spherical source of radius R and source mass parameter M . Retain the finite-domain law with the spherical reference volume and its physical surface readout:

$$V_{\text{ref}} = \frac{4\pi R^3}{3}, Q_{\Omega}^2 = 24\pi GM V_{\text{ref}} = 32\pi^2 GM R^3. \quad (6)$$

$$Q_{\Omega} = 4\pi R^2 S(R). \quad (7)$$

The reference volume is the one used in this source normalization. It is not silently replaced by an integral over an unknown proper radial measure. The equality in equation (7) is the retained source-to-surface correspondence. It is additional physical content beyond naming a charge Q_{Ω} .

Square equation (7) and compare it with equation (6):

$$16\pi^2 R^4 S(R)^2 = 32\pi^2 GM R^3, S(R)^2 = \frac{2GM}{R}. \quad (8)$$

$$A_R = 1 - \frac{2GM}{Rc^2} = 1 - \frac{R_s}{R}, R_s \equiv \frac{2GM}{c^2}. \quad (9)$$

At this stage R_s is a source-charge length. Its interpretation in the exterior solution will follow later. Equation (9) fixes a value at an actual source boundary; it does not yet prescribe $A(r)$ on intermediate vacuum screens. The family is restricted to $R > R_s$, and M is used consistently as the total source mass-energy parameter.

3 The physical gradient and the conserved current

Retain the material proper-time action, with m_p the probe rest mass:

$$I_p = -m_p c^2 \int \beta dT, L_p = -m_p c^2 \beta. \quad (10)$$

In the equatorial plane, with dots denoting d/dT , differentiation of this Lagrangian gives

$$p_r = \frac{m_p \zeta^2 \dot{r}}{\beta}, p_\phi = \frac{m_p r^2 \dot{\phi}}{\beta}. \quad (11)$$

$$E = p_r \dot{r} + p_\phi \dot{\phi} - L_p = \frac{m_p c^2 A}{\beta}. \quad (12)$$

For a held probe, $\beta = \alpha$, so $E_{\text{hold}} = m_p c^2 \alpha$. A quasistatic displacement $d\ell = \zeta dr$ requires local support force F_{hold} ; its work referred to T is $\alpha F_{\text{hold}} d\ell$. Equating this with dE_{hold} gives

$$a_{\text{hold}} = \frac{F_{\text{hold}}}{m_p} = \frac{c^2}{\zeta} \frac{d \ln \alpha}{dr}. \quad (13)$$

The outward redshifted support acceleration is therefore

$$J \equiv \alpha a_{\text{hold}} = \frac{c^2}{\zeta} \frac{d\alpha}{dr} = \frac{c^2}{2\sqrt{H(A)}} \frac{dA}{dr}. \quad (14)$$

Define $d\chi = \alpha d\ell = \sqrt{H(A)} dr$. The same quantity can be written as the physical burden-gradient current retained from 8c:

$$J = -\frac{1}{2} \frac{dS^2}{d\chi}. \quad (15)$$

Appendix A checks the connection of $d\chi$ to the current-adapted radial increment for arbitrary H . It does not set $d\chi = dr$ in advance. The relation between force and held energy above supplies an operational meaning for the gradient.

We now retain 8c's exterior current balance [1, equations P8.65–P8.67]: the flux of this physical-gradient current through any complete spherical vacuum screen is the same source charge.

$$\oint_{\Sigma_r} J \cdot dA_\Sigma = 4\pi GM, r^2 J = GM. \quad (16)$$

Equation (16) is a stated SET field-response law. It is not derived merely from constant source output or from the algebraic invariant. J has units of acceleration and its area integral has units m^3/s^2 ; it is distinct from Q , whose units are m^3/s .

4 Consistency across spherical sources selects the response

Combine the physical gradient in equation (14) with the complete-screen balance. Before choosing any exterior profile, this gives

$$\frac{r^2 c^2}{2\sqrt{H(A)}} \frac{dA}{dr} = GM. \quad (17)$$

Separate the variables and integrate from the physical source surface R to the unloaded limit at infinity:

$$\frac{dA}{\sqrt{H(A)}} = \frac{R_s}{r^2} dr, \quad (18)$$

$$\int_{A_R}^1 \frac{da}{\sqrt{H(a)}} = R_s \int_R^\infty \frac{dr}{r^2} = \frac{R_s}{R}. \quad (19)$$

Independently, the source-surface relation in equation (9) says $R_s/R = 1 - A_R$. Agreement between the two requires

$$\int_{A_R}^1 \frac{da}{\sqrt{H(a)}} = 1 - A_R. \quad (20)$$

Require equation (20) for a continuous family of admissible spherical sources with different compactness, using the same function H . Differentiate this family identity with respect to its boundary value A_R :

$$-\frac{1}{\sqrt{H(A_R)}} = -1, H(A_R) = 1. \quad (21)$$

The fundamental theorem of calculus applies because H is positive and continuous. If the family spans an interval $A_{min} \leq A_R < 1$, this fixes H throughout that interval. It consequently fixes the radial conversion in the exterior of every family member whose exterior capacity lies there:

$$H(A) = 1, \zeta^2 = \frac{1}{A}. \quad (22)$$

The derivative in equation (21) compares different physical source configurations. It does not move one source's boundary through vacuum while keeping its domain volume and Q_Q unchanged. The source law is required to hold at each actual surface, with the corresponding mass and radius.

The selection depends on the range of allowed sources. It does not establish H below the smallest capacity covered by that family. Extending it arbitrarily close to $A=0$ requires admitting an appropriately compact supported family; source stability is not established by this exterior argument. One isolated boundary value is insufficient to select an arbitrary function. Appendix C gives an explicit control of that distinction.

5 The exterior burden and the recovered carrier relation

With the response selected, equation (17) becomes an ordinary differential equation for the exterior clock capacity:

$$\frac{dA}{dr} = \frac{2GM}{c^2 r^2}, A(\infty) = 1. \quad (23)$$

Its integration gives the burden profile and the interval:

$$A(r) = 1 - \frac{2GM}{rc^2}, S(r)^2 = \frac{2GM}{r}. \quad (24)$$

$$c^2 d\tau^2 = A c^2 dT^2 - \frac{dr^2}{A} - r^2 d\Omega^2. \quad (25)$$

This interval is mathematically the Schwarzschild exterior. Here it has followed from the retained source, physical-current and universal-response premises. The argument did not start by imposing an Einstein field equation or the metric-completion action of derivation 13. A free-fall coordinate representation follows in Appendix A after the general radial conversion has been kept explicit.

The spherical domain of the stationary description is $A > 0$. In the extended selected exterior the locus $A = 0$ is $r = R_s$. This limiting statement does not provide a source-interior solution or a dynamical formation calculation.

The conserved current can now be expressed directly through the solved burden:

$$J = \frac{GM}{r^2}, S^4 = \frac{4G^2 M^2}{r^2}, \quad (26)$$

$$J = \frac{S^4}{4GM} = \kappa_M \left(\frac{S^2}{c^2} \right)^2, \kappa_M = \frac{c^4}{4GM}. \quad (27)$$

Equation (27) is the carrier relation assumed in 8c, with exactly its coefficient [1, P8.85]. It is an output in this route. The physical-gradient current and its conserved complete-screen charge remain inputs. Thus the change concerns which relation selects the solution; it does not produce a different spherical orbit.

The later areal-gradient expression follows as well:

$$-\frac{r^2}{2} \frac{dS^2}{dr} = GM. \quad (28)$$

Before H was fixed, the physical gradient contained $1/\sqrt{H}$. Starting only from equation (28) would select the burden profile while leaving the radial conversion unresolved. That distinction is why the physical-current premise was stated explicitly in Section 3.

6 The normalized readout and signal propagation

Define the spherical screen readout from the solved local field. It is distinguished from the value assigned to the material source domain:

$$Q_0(r) = 4\pi r^2 S(r) = 4\pi \sqrt{2GM} r^3. \quad (29)$$

$$\frac{Q_0(r)^2}{24\pi V_{\text{ref}}(r)} = \frac{rS(r)^2}{2} = GM, V_{\text{ref}}(r) = \frac{4\pi r^3}{3}. \quad (30)$$

The normalized charge is constant. The unsquared screen readout Q_0 grows as $r^{3/2}$; it is not a radially conserved volume discharge. The equality to the finite-domain source charge now holds on exterior screens as a property of the solved field.

A held observer uses $d\tau_{\text{hold}} = \alpha dT$. Keeping the same spatial and screen normalization, write $Q_* d\tau_{\text{hold}} = Q_0 dT$. Consequently

$$Q_*(r) = \frac{Q_0(r)}{\alpha(r)} = 4\pi r^{3/2} \sqrt{\frac{2GM}{1 - 2GM/(rc^2)}}. \quad (31)$$

This is the observer-normalized expression used in the earlier algebraic checks. It changes the clock used to report the readout; it does not increase the physical source charge. Its divergence as $A \rightarrow 0$ concerns this held-clock normalization, whose stationary observer description reaches its boundary there.

For radial light, set $d\tau = 0$ in equation (25):

$$\left| \frac{dr}{dT} \right| = cA, d\ell = \frac{|dr|}{\alpha}. \quad (32)$$

$$\frac{d\ell}{dT} = c\alpha, \frac{d\ell}{d\tau_{\text{hold}}} = c. \quad (33)$$

The clock and propagation factors cancel in the local light-speed measurement. The reference-clock signal travel time retains the radial response. More generally, before selection, $|dr/dT| = cA/\sqrt{H}$ and the same local construction still gives c . A local speed of light and the held Q_* expression alone therefore did not select $H = 1$.

7 Conserved probe energy and angular momentum

For equatorial motion in the selected field, the invariant is

$$\beta^2 = A - \frac{\dot{r}^2}{A c^2} - \frac{r^2 \dot{\phi}^2}{c^2}. \quad (34)$$

The action in equation (10) has no explicit T or ϕ dependence. The corresponding conserved quantities follow directly from its derivatives:

$$p_r = \frac{m_p \dot{r}}{A\beta}, p_\phi = \frac{m_p r^2 \dot{\phi}}{\beta} \equiv m_p j. \quad (35)$$

$$E = p_r \dot{r} + p_\phi \dot{\phi} - L_p = \frac{m_p c^2 A}{\beta} \equiv m_p c^2 \varepsilon. \quad (36)$$

In the energy calculation, the velocity terms sum to $c^2(A - \beta^2)$ by equation (34). This gives the final equality without an additional force law. The constants are thus

$$\frac{dT}{d\tau} = \frac{\varepsilon}{A}, \frac{d\phi}{d\tau} = \frac{j}{r^2}. \quad (37)$$

Divide the interval by $d\tau^2$ and substitute these two constants:

$$c^2 = \frac{c^2 \varepsilon^2}{A} - \frac{1}{A} \left(\frac{dr}{d\tau} \right)^2 - \frac{j^2}{r^2}. \quad (38)$$

$$\left(\frac{dr}{d\tau} \right)^2 = c^2 \varepsilon^2 - A \left(c^2 + \frac{j^2}{r^2} \right). \quad (39)$$

The Newtonian limit is also transparent. Expand the same Lagrangian for small $GM/(rc^2)$ and low speed, with $v^2 = \dot{r}^2 + r^2 \dot{\phi}^2$:

$$L_p = -m_p c^2 + \frac{m_p v^2}{2} + \frac{m_p GM}{r} + \dots. \quad (40)$$

At leading order the potential energy is $-m_p GM/r = -m_p S^2/2$. For perihelion, retain the full action and radial integral above. The weak-field expansion is a consistency check, not a replacement for the relativistic motion law.

8 The exact exterior orbit equation

Let $w = 1/r$ and use a prime in this section for $d/d\varphi$. Since $dr/d\varphi = -w'/w^2$ and $d\varphi/d\tau = jw^2$,

$$\frac{dr}{d\tau} = -jw'. \quad (41)$$

Substitute this into equation (39), with $A = 1 - 2GMw/c^2$:

$$j^2(w')^2 = c^2\varepsilon^2 - \left(1 - \frac{2GMw}{c^2}\right)(c^2 + j^2w^2). \quad (42)$$

Expand the product so that the origin of each term remains visible:

$$j^2(w')^2 = c^2(\varepsilon^2 - 1) + 2GMw - j^2w^2 + \frac{2GMj^2w^3}{c^2}. \quad (43)$$

Differentiate with respect to φ :

$$2j^2w'w'' = 2GMw' - 2j^2ww' + \frac{6GMj^2w^2w'}{c^2}. \quad (44)$$

Where $w' \neq 0$, divide by $2j^2w'$. The smooth second-order equation extends to isolated turning points by continuity:

$$w'' + w = \frac{GM}{j^2} + \frac{3GMw^2}{c^2}. \quad (45)$$

The additional term comes from the cubic product in equation (43). Its coefficient has not been fitted to an observed advance. Equation (45) is exact for timelike test motion in the selected spherical exterior. Circular motion satisfies the corresponding radial Euler–Lagrange equation; the turning-point division is unnecessary in that special case.

The response selection matters to the orbit. If the source-energy route fixes the same $A(r)$ but leaves H free, the calculation instead starts from $j^2H(w)(w')^2$ on the left of equation (42). Its derivative gives

$$Hw'' + \frac{1}{2} \frac{dH}{dw} (w')^2 = \frac{GM}{j^2} - w + \frac{3GMw^2}{c^2}. \quad (46)$$

This equation reduces to equation (45) when $H = 1$. Section 10 follows a competing response under both choices of retained field relation, rather than assuming that a different radial coefficient merely relabels the same orbit.

9 From the orbit equation to perihelion advance

Define the gravitational length $\ell_g = GM/c^2$ and the Newtonian angular-momentum scale $p_0 = j^2/(GM)$. At zeroth order, equation (45) gives an ellipse:

$$w_0 = \frac{1 + e \cos \varphi}{p_0}, \quad w_0'' + w_0 = \frac{1}{p_0}. \quad (47)$$

Write $w = w_0 + \delta w$ and retain first order in ℓ_g/p_0 . Substitution into the relativistic term gives

$$\delta w'' + \delta w = 3 \ell_g w_0^2. \quad (48)$$

$$3 \ell_g w_0^2 = \frac{3 \ell_g}{p_0^2} \left(1 + \frac{e^2}{2} + 2e \cos \varphi + \frac{e^2}{2} \cos 2\varphi \right). \quad (49)$$

The constant and double-frequency terms give bounded corrections. The $\cos \varphi$ term is resonant with the unperturbed orbit. The useful identities are

$$\left(\frac{d^2}{d\varphi^2} + 1 \right) (\varphi \sin \varphi) = 2 \cos \varphi, \quad (50)$$

$$\left(\frac{d^2}{d\varphi^2} + 1 \right) \cos 2\varphi = -3 \cos 2\varphi. \quad (51)$$

One particular solution is therefore

$$\delta w = \frac{3 \ell_g}{p_0^2} \left(1 + \frac{e^2}{2} + e\varphi \sin \varphi - \frac{e^2}{6} \cos 2\varphi \right). \quad (52)$$

Homogeneous terms change the fitted eccentricity and initial phase. To identify the secular advance, compare the resonant term with

$$\frac{e}{p_0} \cos[(1 - \delta)\varphi] \approx \frac{e}{p_0} \cos \varphi + \frac{e\delta}{p_0} \varphi \sin \varphi. \quad (53)$$

$$\delta = \frac{3 \ell_g}{p_0}, \quad \Delta\varphi = \frac{2\pi}{1 - \delta} - 2\pi \approx \frac{6\pi \ell_g}{p_0}. \quad (54)$$

At leading order $p_0 = a(1 - e^2)$, giving the perihelion advance per radial period:

$$\Delta\varphi = \frac{6\pi GM}{a(1 - e^2)c^2} + O\left(\frac{G^2 M^2}{a^2(1 - e^2)^2 c^4}\right). \quad (55)$$

The equality between p_0 and the turning-point semilatus rectum is a leading-order identification. Section 11 keeps those quantities separate for the exact orbital check.

10 A competing response followed to its conclusion

Take the response previously retained in the source-energy tests:

$$H(A) = 1 - b(1 - A)^2, b = \frac{1}{2}. \quad (56)$$

It is positive on $0 < A \leq 1$, has the unloaded normalization, and differs from unity only at second order in the weak-field compactness. Earlier energy tests admitted it within their constrained spherical action. We now ask whether it also satisfies the stronger physical-current and source-family correspondence used here. There are two different calculations.

First, keep the inverse-radius profile obtained from the source-energy action. Equation (14) then gives

$$A = 1 - \frac{R_s}{r}, r^2 J = \frac{GM}{\sqrt{H(A)}}. \quad (57)$$

The source charge of that action remains constant, but its physical work-current flux does not equal the stipulated $4\pi GM$ on each screen. At $r = 2R_s$, $A = 1/2$, $H = 7/8$, and $r^2 J / (GM) = 1.069044968$. This version fails equation (16).

Second, enforce the physical-current law and solve for the profile rather than retaining the previous one. Put $x = 1 - A$ and $y = R_s / r$. Equation (17) becomes

$$\frac{dx}{dy} = \sqrt{1 - bx^2}, x(0) = 0. \quad (58)$$

Integration and inversion give

$$\frac{\arcsin(\sqrt{b}x)}{\sqrt{b}} = y, x(y) = \frac{\sin(\sqrt{b}y)}{\sqrt{b}}. \quad (59)$$

This solution conserves the stipulated physical current. At a source boundary, however, the source law requires $x(y_R) = y_R$. For the positive compactness values below, equation (59) gives $x < y$. No orbital velocity is used in the comparison.

R_s / R	Source law $S(R)^2 / c^2$	Alternative $S(R)^2 / c^2$	Difference in S^2
0.05	0.050000	0.049990	-0.021%
0.20	0.200000	0.199334	-0.333%
0.50	0.500000	0.489648	-2.070%
0.80	0.800000	0.758011	-5.249%

The competing response cannot retain both premises for the same source family. Altering the charge at each radius to restore the boundary number would change the specified source. The failure is a consistency result for this branch, not an observational exclusion of every possible alternative to SET.

11 Numerical checks and the Mercury benchmark

For an exact orbital check, define the turning-point semilatus rectum p and eccentricity e by $r_p = p/(1+e)$ and $r_a = p/(1-e)$. Let $P = p/\ell_g$. Requiring equation (42) to vanish at both turning points gives

$$\frac{j^2}{GM \ell_g} = \frac{P^2}{P-3-e^2}, \epsilon^2 = \frac{(P-2)^2 - 4e^2}{P(P-3-e^2)}. \quad (60)$$

These expressions follow by subtracting the two turning-point equations to solve for j^2 , then substituting back for ϵ^2 . They imply $p_0 = p/[1 - (3+e^2)/P]$, so $p_0 = p$ only at leading weak-field order.

Parameterize $w = (1 + e \cos \xi)/p$. Substitution of equation (60) into the expanded radial integral factorizes it as

$$(w')^2 = \frac{e^2 \sin^2 \xi}{p^2} \left[1 - \frac{2(3 + e \cos \xi)}{P} \right]. \quad (61)$$

Cancel the common factor from $(dw/d\xi)^2$. For bound orbits with $P > 6 + 2e$, the exact advance is

$$\Delta\varphi = 2 \int_0^\pi \left[\left(1 - \frac{2(3 + e \cos \xi)}{P} \right)^{-1/2} - 1 \right] d\xi. \quad (62)$$

Numerical quadrature and direct integration of equation (45) give the same advance. As a control, retaining the same inverse-radius profile with the alternative H inserts $\sqrt{H(A(\xi))}$ into the total-angle integrand. At $P = 12$ and $e = 0.2$, the two responses give:

Response	Advance from quadrature rad	Advance from orbit integration rad
$H = 1$	2.6100034270	2.6100034270
$H = 1 - (1 - A)^2/2$	2.5463519904	2.5463519904

Both computations use the same areal turning radii and source charge. The different accumulated azimuth is not removable by relabeling the radial coordinate. This compact orbit is a mathematical check, not a new astronomical measurement. The alternative row belongs to the first branch in Section 10, which keeps the inverse-radius profile.

The Mercury scale calculation

Use the solar gravitational parameter and constants listed by NASA JPL [4], together with its approximate J2000 Mercury elements [5]:

$$G M_{\odot} = 1.3271244004127942 \times 10^{20} \text{ m}^3/\text{s}^2, \quad (63)$$

$$c = 299792458 \text{ m/s}, 1 \text{ au} = 149597870700 \text{ m}, \quad (64)$$

$$a = 0.38709927 \text{ au}, e = 0.20563593. \quad (65)$$

The orbital scale and small expansion parameter are

$$p = a(1 - e^2) = 5.546046913 \times 10^{10} \text{ m}, \quad (66)$$

$$\frac{G M_{\odot}}{p c^2} = 2.662482056 \times 10^{-8}. \quad (67)$$

Equation (55) then gives the leading advance per orbit. To express it per Julian century, use 36525 days per century [4] and the listed sidereal period 87.969 days [6].

$$\Delta \phi_{\text{orbit}} = 5.018660440 \times 10^{-7} \text{ rad} = 0.1035173023 \text{ arcsec}, \quad (68)$$

$$\Delta \phi_{\text{century}} = 0.1035173023 \frac{36525}{87.969} \text{ arcsec} = 42.98070 \text{ arcsec}. \quad (69)$$

The exact spherical quadrature, evaluated with the same adopted turning-point p and e , gives 42.9807082 arcseconds per Julian century. The extra digits record agreement between calculations; the approximate orbital inputs do not support that precision as an empirical determination.

This recovers the usual approximately 43 arcseconds per century spherical relativistic contribution. The result is the same as Schwarzschild test-body dynamics. It is a numerical benchmark for the derivation, rather than a new fit to Mercury observations or a calculation of the full Solar System precession.

The verification also checks the source-family integral against an independently integrated field equation, differentiates the general orbital integral with arbitrary radial response, and checks the current-adapted transformation without fixing H . The accompanying research calculation passed 305 exact rational identity checks. The numerical field integration agreed with equation (59) to below 6×10^{-13} in the tested dimensionless burden. These are algebra and implementation checks, not measurement uncertainties.

12 What this changes relative to 8c and 13

Derivation 8c supplied two descriptions of the same current: a physical burden gradient and a carrier proportional to the square of S^2/c^2 . The carrier assumption, together with current conservation, directly selected an inverse-radius burden profile. The physical-gradient relation then calibrated the radial conversion.

Here the physical-gradient current and its complete-screen charge are retained, while a common response is required to agree with the source-surface normalization across a continuous family. That selects $H=1$. The inverse-radius field and the carrier relation follow. This is a change in the logical dependencies of the spherical construction, not a new numerical perihelion prediction.

Element	Derivation 8c	Derivation 14
Capacity invariant and probe action	Retained	Retained
Physical-gradient current and charge	Assumed	Retained as a field law
Carrier proportional to the square of burden fraction	Assumed to select the profile	Recovered from the solution
Additional selection structure	Carrier relation and unloaded calibration	Universal $H(A)$ and continuous source family
Exterior orbit	Standard spherical relativistic orbit	Same orbit

Derivation 13 selected the spherical action through common evolution and a local metric completion. The present route does not use that metric action to select the stationary exterior. It consequently does not inherit all of that completion's dynamical content merely because the exterior solution agrees. Appendix B records the connection to the separately developed source-energy action.

The result established here is a conditional spherical closure: source-surface normalization, physical-current conservation and a universal local response jointly determine the radial conversion and burden profile over the family's capacity interval. The material action then determines the orbit. The scalar invariant alone does not establish the current law, the allowed response arguments, or the range of physical source configurations.

This distinction preserves the useful advance. A carrier relation that was previously supplied can now be checked as a consequence of the source correspondence. Extending the result to arbitrary field evolution, nonspherical sources or galaxy effective areas requires a further construction; the present calculation makes no such extension.

Appendix A The current adapted increment before selection

The physical-current interpretation can be checked without assuming the final radial coefficient. Write $h = \sqrt{H}$ in this appendix. This h is neither the burden fraction nor a new field. Equation (5) reads

$$c^2 d\tau^2 = A c^2 dT^2 - \frac{h^2}{A} dr^2 - r^2 d\Omega^2. \quad (70)$$

For inward marginal-energy radial free fall, the probe energy has $\varepsilon = 1$ and $j = 0$. Equations (12) and (70) give

$$\frac{dT}{d\tau} = \frac{1}{A}, \quad \frac{dr}{d\tau} = -\frac{c\sqrt{1-A}}{h} = -\frac{S}{h}. \quad (71)$$

Define a new clock increment in this stationary radial domain:

$$dt_f = dT + \frac{Sh}{Ac^2} dr. \quad (72)$$

Its coefficient depends only on r , so it integrates locally to a time-coordinate transformation. Along the free-fall trajectories above,

$$\frac{dt_f}{d\tau} = \frac{1}{A} - \frac{S^2}{Ac^2} = 1. \quad (73)$$

Substitute $dT = dt_f - Sh dr / (Ac^2)$ into equation (70). The cross term is $-2Sh dt_f dr$ and the radial square is $-h^2 dr^2$, yielding

$$c^2 d\tau^2 = c^2 dt_f^2 - (h dr + S dt_f)^2 - r^2 d\Omega^2. \quad (74)$$

The radial increment on a t_f slice is $d\chi = h dr = \alpha d\ell$. Its physical burden gradient is

$$-\frac{1}{2} \frac{dS^2}{d\chi} = \frac{c^2 \alpha \alpha_r}{h} = \frac{c^2 \alpha_r}{\zeta} = J. \quad (75)$$

All these identities hold for arbitrary positive H in the stated domain. They identify the current-adapted reading used in Section 3; they do not select $h = 1$. After the source-family argument sets $H = 1$, equation (74) becomes the usual ingoing Painlevé–Gullstrand representation of the solved exterior. Its algebraic appearance does not establish a literal emitted fluid.

For this unit-energy free-fall congruence, the shell volume is $dV_f = 4\pi r^2 h dr$. It agrees with the source reference-shell volume $4\pi r^2 dr$ after selection. The equality was not inserted to obtain equation (21), and it is not asserted for arbitrary observer congruences.

Appendix B Relation to the source energy construction

A separate research route sought a static action from source-energy accounting within a restricted first-gradient spherical class [7]. Its retained action was

$$I_{g,\text{static}} = -\frac{c^4}{G} \int dT dr \alpha P(z), z = \frac{r \alpha_r}{\alpha}, \quad (76)$$

$$P(z) = 1 + z - \sqrt{1 + 2z}. \quad (77)$$

This appendix checks agreement with that action; the family selection in Section 4 does not depend on it. The action class and its source-surface energy correspondence were physical inputs in that earlier research, not consequences of the scalar budget alone.

With $h_c = \ln \alpha$, the Euler–Lagrange equation obtained by variation of h_c is

$$\frac{d}{dr} [r \alpha P_z] = \alpha P. \quad (78)$$

Expand the derivative, using $\alpha_r = \alpha z / r$:

$$(1 + z) P_z + r P_{zz} z_r - P = 0. \quad (79)$$

Define $K(z) = (1 + z) P_z - P$. Its derivative is $K_z = (1 + z) P_{zz}$. Equation (79) then verifies the first integral directly:

$$\frac{d}{dr} [r \alpha K] = \alpha (1 + z) [K + r P_{zz} z_r] = 0. \quad (80)$$

For the function in equation (77), define $v = \sqrt{1 + 2z}$. Then

$$P_z = 1 - \frac{1}{v}, P_{zz} = \frac{1}{v^3}, K = \frac{z}{v}. \quad (81)$$

Normalize the constant by the isolated source strength:

$$GM = c^2 r \alpha K = \frac{c^2 r^2 \alpha_r}{v}. \quad (82)$$

The same field equation gives $r z_r = -z(1 + 2z)$, so $v_r = -z v / r$. Together with $\alpha_r = \alpha z / r$, this gives $(\alpha v)_r = 0$. The unloaded condition sets $\alpha v = 1$. Equation (82) therefore integrates to $\alpha^2 = 1 - 2GM / (r c^2)$.

Source current and physical work in the energy route

The first integral of the source action and the probe's physical work readout define two currents before their radial responses are identified:

$$J_{\text{source}} = \frac{c^2 \alpha_r}{v}, J_{\text{work}} = \frac{c^2 \alpha_r}{\zeta}. \quad (83)$$

In the isolated vacuum exterior $\alpha v = 1$, hence

$$J_{\text{source}} = \sqrt{H} J_{\text{work}}. \quad (84)$$

Requiring the physical work-current to carry the same complete-screen charge selects $\zeta = v = 1/\alpha$ wherever the source gradient is nonzero. This is the same radial response selected by Section 4. The source-family route makes that selection without first choosing equation (76).

Energy conservation by itself did not enforce this identification. For an arbitrary positive radial conversion, quasistatic lifting satisfies

$$J_{\text{work}} d\ell = \frac{c^2 \alpha_r}{\zeta} (\zeta dr) = c^2 d\alpha. \quad (85)$$

The force and physical distance change reciprocally. Integrating equation (85) between two fixed clock capacities gives the same lifting energy for different H , even though their radial signal times and orbital angles can differ. Previous coupled spherical source-motion checks [7] also preserved total energy for the surviving alternative when its response-dependent force terms were retained.

For example, retaining the common profile $A = 1 - R_s/r$, an outward lift from $2R_s$ to $10R_s$ requires $0.2415765169 m_p c^2$ for both $H=1$ and $H=1 - (1-A)^2/2$. A radial light round trip between those radii, timed by the held emitter at $2R_s$, instead takes

$$\Delta \tau_{\text{em}} = 2 \sqrt{A(2R_s)} \int_{2R_s}^{10R_s} \frac{\sqrt{H(A(r))}}{cA(r)} dr. \quad (86)$$

Radial response	Emitter clock round trip in units of R_s divided by c
$H = 1$	14.421053296
$H = 1 - (1 - A)^2/2$	14.210116208

These are internal comparison calculations with fixed source charge and areal endpoints. They show why conservation of lifting energy alone could not replace the physical-current premise. They do not establish the alternatives as complete theories, or claim observational support for either response.

Appendix C Why the continuous family matters

For compactness $x = 1 - A_R$, define the integrated response

$$F(x) = \int_0^x \frac{dt}{\sqrt{H(1-t)}}. \quad (87)$$

The source correspondence requires $F(x) = x$. At one chosen surface $x = x_0$, this is one integral constraint on a function. It does not force the integrand to equal one. To see this explicitly, set

$$F(x) = x + \eta x^2(x - x_0), x_0 = \frac{1}{2}, \eta = \frac{1}{5}. \quad (88)$$

$$F'(x) = 1 + \eta(3x^2 - 2x_0x), H(1-x) = \frac{1}{[F'(x)]^2}. \quad (89)$$

The derivative stays positive on $0 \leq x \leq 1$. The unloaded normalization is $H(1) = 1$, while $F(x_0) = x_0$ gives exact agreement at the selected surface. Nevertheless,

$$H(1 - x_0) = \frac{1}{(1 + \eta x_0^2)^2} = 0.9070294785 \neq 1. \quad (90)$$

This example is a control of the mathematical family argument. It is not claimed to satisfy every source-energy or capacity restriction investigated in the separate dynamical work. Its purpose is to show why agreement at a single boundary would not establish the functional selection in Section 4.

Across a continuous interval, by contrast, $F(x) = x$ implies $F'(x) = 1$, which fixes $H = 1$ there. This inference requires the same function in each system. If the response were independently allowed to depend on source size or other variables, differentiation across sources would also differentiate those arguments, and equation (21) would no longer follow.

The physical question attached to the proof is therefore precise: does the stationary spherical response belong to the stipulated universal local class, and does the physical-gradient current obey the retained complete-screen law? The derivation establishes the consequences of those SET premises and exposes where they enter.

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